



Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE342.2

Experiment No.: 06

Experiment Name: Inter-vlan Configuration with static Routing, DNS and WEB server Configuration.

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Batch: 60

❑ Experiment No: 06

❑ Experiment Name: Inter-VLAN Configuration with static Routing, DNS and Web Server.

❑ Introduction: Network segmentation through virtual Local Area networks (VLANs) isolates broadcast domains improving security and network efficiency. Because VLANs operate at layer-2 and layer-3 routing is required for inter-vlan communication. This lab utilizes a "Router-on-a-stick" configuration utilizing a single physical router interface divided into multiple sub-interfaces acting as default gateways for each vlan. Static routing is then implemented to route traffic across distinct networks, allowing end-devices to resolve domain names via a dedicated DNS server and access HTTP services hosted on a remote web server.

❑ Objective:

- Configure VLAN 10 and VLAN 20 on layer 2 switches and assign access ports.
- Establish 802.1Q trunk links between switches and the inter-vlan router.
- Configure a router-on-a-stick topology for inter-vlan routing.

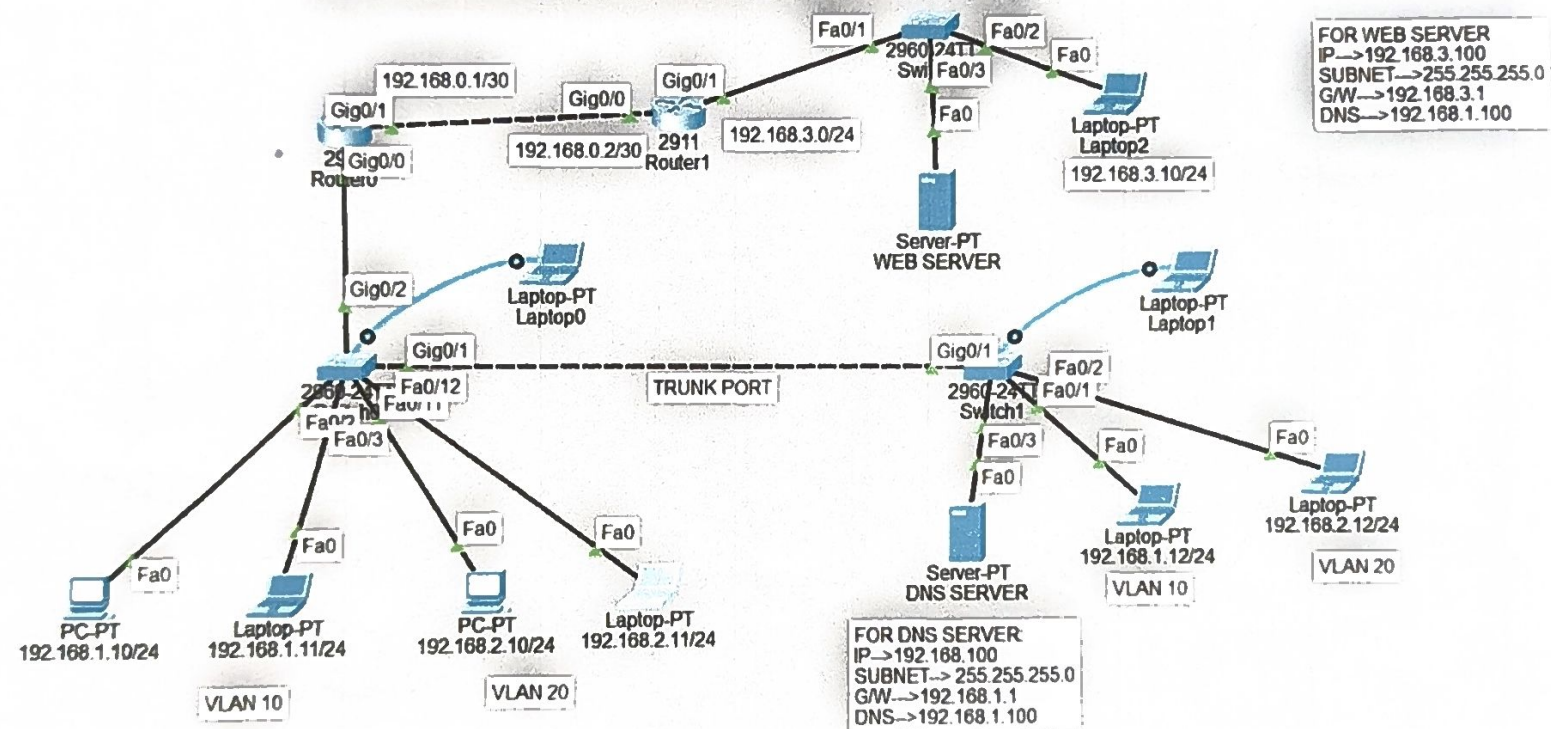
- Implement static routing between Router 0 and Router 1 to enable Point-to-Point network communication.
- Deploy and verify DNS and web server functionalities across the routed network.

Network Addressing Table:

Device	Interface	IP Address	Subnet mask	Default gateway
Router 0	Gig 0/1	192.168.0.1	255.255.255.252	N/A
Router 0	Gig 0/2.10	192.168.1.1	255.255.255.0	N/A
Router 0	Gig 0/2.20	192.168.2.1	255.255.255.0	N/A
Router 1	Gig 0/0	192.168.0.2	255.255.255.252	N/A
Router 1	Gig 0/1	192.168.3.1	255.255.255.0	N/A
DNS server	Through VLAN 10 Fa 0/3	192.168.1.100	255.255.255.0	192.168.1.1
WEB server	Remote switch Fa 0/3	192.168.3.100	255.255.255.0	192.168.3.1

- For the end-device IP configuration we need to open GUI of every end-devices and assign IP and DNS statically.

INTER VLAN WITH STATIC ROUTING CONFIGURATION AND DNS WITH WEB SERVER



Time: 00:03:11

Realtime Simulation

(Select a Device to Drag and Drop to the Workspace)

Scenario 0
New Delete
Toggle PDU List Window

Time	List	Source	Destination	Type	Color	Time (Sec)	Periodic	Num	Edit	Delete
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Step by Step Configuration:

• VLAN Creation and Port Access:

Switch > Enable

Switch# configure terminal

Switch(config)# VLAN 10

Switch(config)# Name students

Switch(config)# Exit

Switch(config)# VLAN 20

Switch(config)# Name faculties

Switch(config)# Exit

Switch(config)# Interface range fa 0/2-3

Switch(config)# Switchport mode access

Switch(config)# Switchport access vlan 10

Switch(config)# exit

Switch(config)# Interface range fa 0/11-12

Switch(config)# Switchport mode access

Switch(config)# Switchport access vlan 20

Switch(config)# Exit

• Trunk Configuration:

→ on switch0:

Switch(config)# Interface gig 0/1

Switch(config)# Switchport mode trunk

Switch(config)# Switchport trunk allowed vlan all

Switch(config)# Exit

Switch(config)# Interface gig 0/2

```
Switch(config)# Switchport mode access
Switch(config)# Switchport trunk allowed vlan all
Switch(config)# Exit
```

→ on Switch 1:

```
Switch(config)# Interface gig 0/1
Switch(config)# Switchport mode access
Switch(config)# Switchport trunk allowed vlan all
Switch(config)# Exit
```

• Inter-vlan Routing:

```
Router > Enable
Router# Configure terminal
Router(config)# Interface gig 0/2
Router(config)# no shutdown
Router(config)# Interface gig 0/2.10
Router(config)# encapsulation dot1q 10
Router(config)# IP address 192.168.1.1 255.255.255.0
Router(config)# Interface gig 0/2.20
Router(config)# encapsulation dot1q 20
Router(config)# IP address 192.168.2.1 255.255.255.0
Router(config)# Exit.
```

• Static Routing Configuration:

```
→ Router0: Router(config)# Interface gig 0/1
Router(config)# IP address 192.168.0.1 255.255.255.252
Router(config)# no shutdown
Router(config)# Exit
Router(config)# IP route 192.168.3.0 255.255.255.0 192.168.0.2
```

→ Router 1:

Router(config)# interface gig 0/0

Router(config)# IP address 192.168.0.2 255.255.255.252

Router(config)# No shutdown

Router(config)# interface gig 0/1

Router(config)# IP address 192.168.3.1 255.255.255.0

Router(config)# No shutdown

Router(config)# Exit

Router(config)# IP route 192.168.1.0 255.255.255.0 192.168.0.1

Router(config)# IP route 192.168.2.0 255.255.255.0 192.168.0.1

☐ Servers Configuration:

- WEB server: To configure web server need to follow these steps. As it is in the remote router we don't need to assign it to any vlan interface.

IP → 192.168.3.100

Subnet → 255.255.255.0

Gateway → 192.168.3.1

DNS server → 192.168.1.100

Ensure that HTTP/HTTPS services are ON.

- DNS Server: To Configure DNS server for this topology we need to assign it into vlan 10 to access the network of vlan 10.

IP $\rightarrow$ 192.168.1.100

Subnet $\rightarrow$ 255.255.255.0

Gateway $\rightarrow$ 192.168.1.1

DNS $\rightarrow$ 192.168.1.100

$\rightarrow$ Also Turn ON the DNS service from services section.
Add a record mapping your domain [www.Rafin.com]
to 192.168.3.100.

- End-Devices: Assign IPs corresponding to their VLAN Subnets [192.168.1.0/24 or 192.168.2.0/24] with the gateway pointing to the Router's sub-interface IP and DNS set to 192.168.1.100

Verification:

- To verify VLAN and Trunk:

show vlan brief

show interfaces trunk

on switch 0 and switch 1, verify that VLANs are active and assigned to the correct ports.

- Verify Routing Table: To Check the routing Table on the both routers to Confirm static routes are injected correctly.

Show IP route

- End-to-End Connectivity: Open the Command Prompt on a PC in VLAN 10 and Ping the other VLAN 20 PC or Web server to ensure Inter-VLAN and static routing are fully functional.

Ping 192.168.3.100

- DNS and WEB Server: From the same PC in VLAN 10, Open the web Browser application in Interface. Type the domain name configured in the DNS server [Ratin.com] into the URL bar and Press go.